

Phase 3: Data Modeling & Relationships

Doctor Appointment & Consultation System (Healthcare CRM)

❖ Objective

To design and build the **data model** for the Doctor Appointment & Consultation System by creating custom objects, defining relationships, and configuring layouts.

This phase focuses on structuring the CRM data layer to support patient management, appointment scheduling, and prescription handling.

❖ Standard Objects Used

Standard Object	Description
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User	Represents doctors, system admins, and future support users.
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Account / Contact	Optional for extended patient or organization mapping (future use).
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Case	Reserved for handling patient complaints or follow-up cases in later phases.
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❖ Custom Objects Created

The following **custom objects** were created to capture and manage healthcare data efficiently:

Object Label	API Name	Record Name	Data Type	Description
Patient	Patient__c	Patient Name (Text)	Text	Stores patient details and assigned doctor.
Appointment	Appointment__c	Appointment Number (Auto Number: AP-000{0})	Auto Number	Tracks patient appointments and consultation details.
Prescription	Prescription__c	Prescription Number (Auto Number: PR-000{0})	Auto Number	Stores medicines prescribed by doctors.

All objects were configured with:

- *Allow Reports*
- *Allow Activities*

- Track Field History

SETUP > OBJECT MANAGER

Patient

Details

Fields & Relationships

Page Layouts

Lightning Record Pages

Buttons, Links, and Actions

Compact Layouts

Field Sets

Object Limits

Record Types

Related Lookup Filters

Search Layouts

List View Button Layout

Restriction Rules

Details

Description

API Name

Patient__c

Custom

✓

Singular Label

Patient

Plural Label

Patients

Enable Reports

✓

Track Activities

✓

Track Field History

✓

Deployment Status

Deployed

Help Settings

Standard salesforce.com Help Window

EditDelete

Patient_c

Setup

Home

Object Manager

Search Setup

Star Plus Cloud ? Bell Profile

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Scoping Rules

Object Access

Details

Description

API Name

Appointment__c

Custom

✓

Singular Label

Appointment

Plural Label

Appointments

Enable Reports

✓

Track Activities

✓

Track Field History

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Scoping Rules

Object Access

Details

Description

API Name

Prescription__c

Custom

✓

Singular Label

Prescription

Plural Label

Prescriptions

Enable Reports

✓

Track Activities

✓

Track Field History

Deployment Status

Deployed

Help Settings

Standard salesforce.com Help Window

EditDelete

Prescription_c

❖ Custom Fields Created

Patient Object

Field Label	Data Type	Description
Email	Email	Patient's email address.
Phone	Phone	Patient contact number.
Date of Birth	Date	Used for patient age tracking.
Gender	Picklist (Male, Female, Other)	Identifies gender.
Primary Doctor Lookup (User)		Links each patient to their primary doctor.

SETUP > OBJECT MANAGER				
Patient				
Details	Fields & Relationships			
Fields & Relationships	9 Items. Sorted by Field Label			
Page Layouts	Quick Find New Deleted Fields Field Dependencies Set History Tracking			
Lightning Record Pages				
Buttons, Links, and Actions				
Compact Layouts				
Field Sets				
Object Limits				
Record Types				
Related Lookup Filters				
Search Layouts				
List View Button Layout				
Restriction Rules				
Scoping Rules				
Object Access				
	FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD INDEXED
	Created By	CreatedById	Lookup(User)	
	Date of Birth	Date_of_Birth__c	Date	
	Email	Email__c	Email	
	Gender	Gender__c	Picklist	
	Last Modified By	LastModifiedById	Lookup(User)	
	Owner	OwnerId	Lookup(User,Group)	✓
	Patient Name	Name	Text(80)	✓
	Phone number	Phone_number__c	Phone	
	Primary Doctor	Primary_Doctor__c	Lookup(User)	✓

Appointment Object

Field Label	Data Type	Description
Appointment Date	Date/Time	Scheduled date and time for the consultation.
Status	Picklist (Scheduled, Completed, Cancelled)	Represents appointment status.
Consultation Type	Picklist (Online, In-Person)	Defines consultation mode.
Patient	Master-Detail (Patient)	Links appointment with a specific patient.
Doctor	Lookup (User)	Assigns appointment to a doctor.

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Object Access

Fields & Relationships

8 Items, Sorted by Field Label

Q Quick Find

New

Deleted Fields

Field Dependencies

Set History Tracking

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Appointment Date	Appointment_Date__c	Date/Time		
Appointment Name	Name	Auto Number		✓
Consultation Type	Consultation_Type__c	Picklist		
Created By	CreatedById	Lookup(User)		
Doctor	Doctor__c	Lookup(User)		✓
Last Modified By	LastModifiedById	Lookup(User)		
Patient	Patient__c	Master-Detail(Patient)		✓
status	status__c	Picklist		

Prescription Object

Field Label	Data Type	Description
Prescription Date	Date	Date when prescription was issued.
Medication Details	Long Text Area (32000 chars)	Contains prescribed medicine details.
Appointment	Lookup (Appointment)	Links prescription to appointment.
Issued By	Lookup (User)	Doctor who issued the prescription.
Remarks	Text Area (255)	Notes or comments by doctor.

SETUP > OBJECT MANAGER

Prescription

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Related Lookup Filters

Restriction Rules

Scoping Rules

Object Access

Triggers

Flow Triggers

Details

Description

API Name

Prescription__c

Custom

✓

Singular Label

Prescription

Plural Label

Prescriptions

Enable Reports

✓

Track Activities

✓

Track Field History

Deployment Status

Deployed

Help Settings

Standard salesforce.com Help Window

Edit

Delete

❖ Relationships Created

Parent Object	Child Object	Relationship Type	Description
Patient	Appointment	Master-Detail	Appointments inherit patient ownership and sharing.
Appointment	Prescription	Lookup	Each prescription belongs to one appointment.
User (Doctor)	Patient / Appointment / Prescription	Lookup	Links records to responsible doctors.

Schema Builder Verification:

- Opened *Setup* → *Schema Builder* → Dragged **Patient, Appointment, Prescription**.
- Verified **Master-Detail** line (Patient → Appointment) and **Lookup** (Appointment → Prescription).
- Checked Lookup(User) for all doctor relationships.

❖ Page Layouts

Customized **Page Layouts** were configured for each object to enhance usability and display related records.

Patient Layout:

- Fields: Patient Name, Email, Phone, Gender, DOB, Primary Doctor.
- Related Lists: Appointments, Prescriptions.

The screenshot shows a 'New Patient' form in a CRM system. The form is titled 'New Patient' and has a 'Required Information' section. Fields include Patient Name (with a blue border), Email, Phone number, Date of Birth (with a calendar icon), Gender (dropdown menu), and Primary Doctor (with a search icon). The form is set against a background of a CRM interface showing a 'Patients' list with columns for Name and Email. The 'Primary Doctor' field is set to 'N Manoj Bhushan'.

Appointment Layout:

- Fields: Appointment Date, Status, Consultation Type, Patient, Doctor.
- Related Lists: Prescription.

The screenshot shows a 'New Appointment' form overlaid on a CRM interface. The form has a title bar with a close button. Below the title is a section labeled 'Information' with a red asterisk indicating required information. The fields include: 'Appointment Name' (text input), 'Appointment Date' (date and time pickers, with a calendar icon for the date and a clock icon for the time, and a format '12/31/2024'), 'Patient' (searchable dropdown), 'Doctor' (searchable dropdown), 'Status' (dropdown menu with '--None--'), and 'Consultation Type' (dropdown menu with '--None--'). At the bottom are 'Cancel', 'Save & New', and 'Save' buttons. The background shows a sidebar with 'Sales', 'Home', 'Opportunities', 'Leads', and 'Tasks' menus, and a main area with 'Appointments' and 'Recently Viewed' sections.

Prescription Layout:

- Fields: Prescription Date, Medication Details, Appointment, Issued By, Remarks.

The screenshot shows a 'New Prescription' form overlaid on a CRM interface. The form has a title bar with a close button. Below the title is a section labeled 'Information' with a red asterisk indicating required information. The fields include: 'Prescription Name' (text input), 'Owner' (dropdown menu showing 'N Manoj Bhushan'), 'Prescription Date' (date and time pickers, with a calendar icon for the date and a clock icon for the time, and a format '12/31/2024'), 'Medication Details' (text area), 'Appointment' (searchable dropdown), 'Issued By' (searchable dropdown), and 'Remark' (text area). At the bottom are 'Cancel', 'Save & New', and 'Save' buttons. The background shows a sidebar with 'Sales', 'Home', 'Opportunities', 'Leads', and 'Tasks' menus, and a main area with 'Prescriptions' and 'Recently Viewed' sections.

❖ Compact Layouts

Patient Compact Layout: Patient Name, Phone, Email, Primary Doctor.

Appointment Compact Layout: Appointment Date, Status, Consultation Type.

Prescription Compact Layout: Prescription Date, Issued By.

→ Marked each as **Primary Compact Layout**.

❖ Record Types

Two **Record Types** were created for the Appointment object to categorize consultations:

- Online Consultation
- In-Person Visit

Each record type uses separate picklist values to streamline appointment entry.

❖ Profiles and Field-Level Security

Profile	CRUD Permissions	Notes
System Administrator	Read, Create, Edit, Delete	Full access to all records and objects.
Doctor	Read, Create, Edit (no Delete)	Limited access for operational use.

Configured **Field-Level Security (FLS)** so the **Doctor** profile can view and edit required fields while keeping system-critical fields protected.